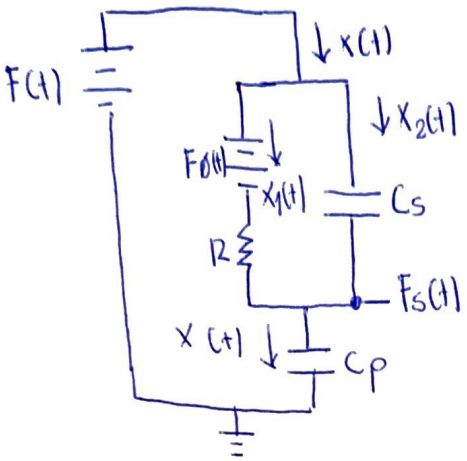
Practica 3: Sistema musculoesqueletico

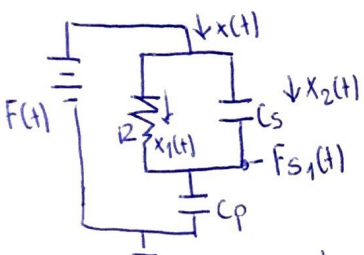


$$x(t) = X_1(t) + X_2(t)$$

$$F_s(t) = F_{s_1}(t) + F_{s_2}(t)$$

Analisis matematico: Superposición

Circuito 1 apagando Fø(t)



Ecuaciones principales

$$X(t) = X_1(t) + X_2(t)$$

$$X(t) = C_p \frac{d\,F_{s_1}(t)}{dt}$$

$$X_1(t) = \frac{F(t) - F_{s_1}(t)}{R}$$

$$X_2(t) = C_s \frac{d\,[F(t) - F_{s_1}(t)]}{dt}$$

$$C_p \frac{d\,F_{s_1}(t)}{dt} = \frac{F(t) - F_{s_1}(t)}{R} + C_s \frac{d\,[F(t) - F_{s_1}(t)]}{dt}$$

Transformada de Laplace

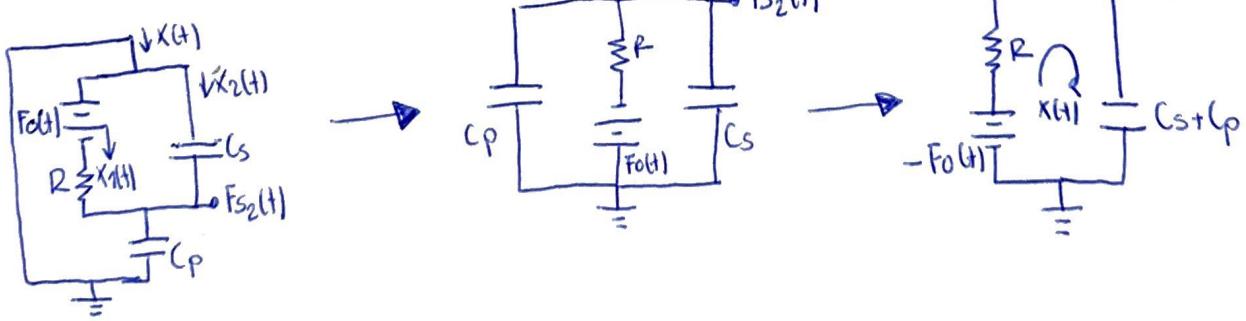
$$C_p s\,F_{s_1}(s) = \frac{F(s) - F_{s_1}(s)}{R} + C_s s\,[F(s) - F_{s_1}(s)]$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s_1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

$$\left[\frac{R[C_p + C_s]s + 1}{R}\right] F_{s_1}(s) = \left[\frac{C_s R s + 1}{R}\right] F(s)$$

$$F_{s_1}(s) = \frac{\frac{C_s R s + 1}{R}}{\left[\frac{R[C_p + C_s]s + 1}{R}\right]} F(s) = \frac{C_s R s + 1}{R[C_p + C_s]s + 1} F(s)$$

Circuito 2 apagando F(t):



Ecuaciones principales

Recordando que: $F_o(t) = A f(t)$

$$-A f(t) = R x(t) + \frac{1}{C_s + C_p} \int x(t)\, dt$$

$$F_{s_2}(t) = \frac{1}{C_s + C_p} \int x(t)\, dt$$

Transformada de Laplace

$$-A f(s) = R x(s) + \frac{X(s)}{[C_s + C_p]s}$$

$$F(s) = \left[R x(s) + \frac{X(s)}{[C_s + C_p]s} \right] \frac{1}{A} = -\left[\frac{R[C_p + C_s]s + 1}{A[C_s + C_p]s} \right] X(s)$$

$$F_{s2}(s) = \frac{X(s)}{[C_s + C_p]s}$$

Procedimiento algebraico

$$\frac{F_{s_2}(s)}{F(s)} = \frac{\frac{X(s)}{[C_s + C_p]s}}{\left[\frac{R[C_s + C_p]s + 1}{A[C_s + C_p]s} \right] X(s)} = -\frac{A}{R[C_s + C_p]s + 1}$$

$$F_{s_2}(s) = -\frac{A}{R[C_s + C_p]s} F(s)$$

Función de transferencia

$$F_s(s) = F_{s_1}(t) + F_{s_2}(t) = \frac{C_s R s + 1}{R[C_p + C_s]s + 1} F(s) - \frac{A}{R[C_p + C_s]s + 1} F(s)$$

$$F_s(s) = \frac{C_s R s + 1 - A}{R[C_p + C_s]s + 1} F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R s + 1 - A}{R[C_p + C_s]s + 1}$$

Error en estado estacionario

$$e(s) = \lim_{s \to 0} F(s)\left(1 - \frac{F_s(s)}{F(s)}\right) = \lim_{s \to 0} \frac{1}{s}\left(1 - \frac{C_s R s + 1 - A}{R[C_p + C_s]s + 1}\right) = A$$

Estabilidad en lazo abierto

$$R[C_p + C_s]s + 1 = 0$$

El sistema es estable

$$\lambda = -\frac{1}{R[C_p + C_s]}$$